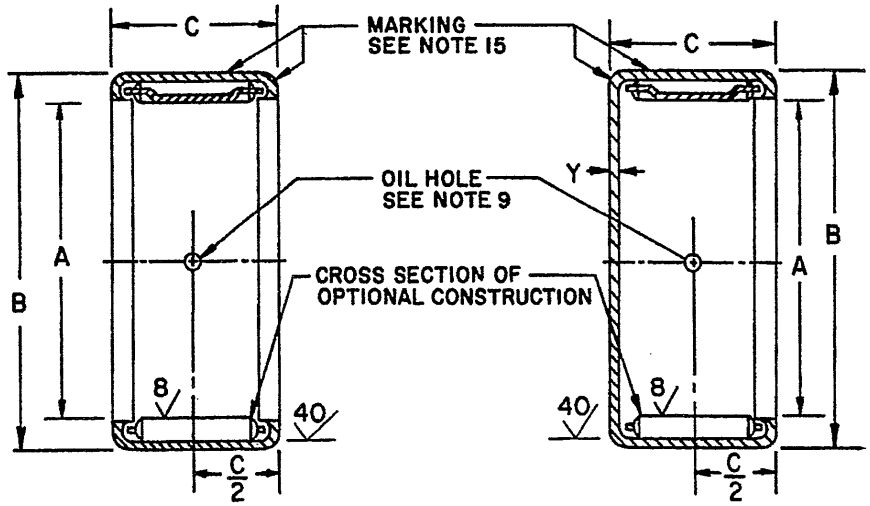
REVIEW ACTIVITIES: ARMY-ME, AV, MI; AF-99; DLA-IS
USER ACTIVITIES: ARMY-AR, GL; NAVY-YD, MC, SH

FED. SUP CLASS
3110

OPEN END
(TYPE J)

CLOSED END
(TYPE MJ)



(NOTE 15) (NOTE 4) (NOTE 6) (NOTE 4) (NOTE 10) (NOTE 11) (NOTE 13)

MS DASH NO.	A BORE			B DIA		C WIDTH +.000 -.010	RECOMMENDED SHAFT DIA		BASIC STATIC LOAD RATING LB	BASIC DYNAMIC LOAD RATING LB	APPROX LIMITING SPEED RPM	Y END THICK-NESS MAX.
	NOM	MIN	MAX	RING	OUTSIDE		MIN	MAX				
MS52141 -1	3/16	.1883	.1892	.3437	11/32	.375	.1872	.1875	217	288	75000	.04
MS " -2	1/4	.2515	.2524	.4380	7/16	.312	.2495	.2500	172	256	57000	.04
MS " -3	1/4	.2515	.2524	.4380	7/16	.438	.2495	.2500	370	464	57000	.04
MS " -4	5/16	.3140	.3149	.5005	1/2	.312	.3120	.3125	215	303	44000	.04
MS " -5	5/16	.3140	.3149	.5005	1/2	.438	.3120	.3125	463	550	44000	.04
MS " -6	3/8	.3765	.3774	.5630	9/16	.375	.3745	.3750	403	487	36000	.04
MS " -7	3/8	.3765	.3774	.5630	9/16	.500	.3745	.3750	692	743	36000	.04
MS " -8	7/16	.4390	.4399	.6255	5/8	.500	.4370	.4375	808	822	30000	.04
MS " -9	1/2	.5015	.5024	.6880	11/16	.500	.4995	.5000	923	894	26000	.04
MS " -10	1/2	.5015	.5024	.6880	11/16	.750	.4995	.5000	1740	1460	26000	.04
MS " -11	9/16	.5640	.5649	.7505	3/4	.500	.5620	.5625	923	881	23000	.04
MS " -12	5/8	.6265	.6274	.8130	13/16	.500	.6245	.6250	1040	951	21000	.04
MS " -13	5/8	.6265	.6274	.8130	13/16	.750	.6245	.6250	1950	1550	21000	.04
MS " -14	11/16	.6890	.6899	.8755	7/8	.750	.6870	.6875	2170	1660	19000	.04
MS " -15	3/4	.7505	.7514	.9995	1	.500	.7495	.7500	1320	1330	17000	.04
MS " -16	3/4	.7505	.7514	.9995	1	.750	.7495	.7500	2570	2230	17000	.04

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P.A. AT Other Cust NAVY-OS AF-11	TITLE BEARING, ROLLER, NEEDLE: DRAWN OUTER RING, CAGED, WITHOUT INNER RING OPEN AND CLOSED END, STANDARD TYPE	MILITARY STANDARD MS52141
PROCUREMENT SPECIFICATION NONE	SUPERSEDES:	SHEET 1 OF 3

DD FORM 1 SEP 87 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. 3110-0587

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(NOTE 15) (NOTE 4) (NOTE 6) (NOTE 4) (NOTE 10) (NOTE 11) (NOTE 13)

MS DASH NO.	A BORE			B DIA		C WIDTH +.000 -.010	RECOMMENDED SHAFT DIA		BASIC STATIC LOAD RATING LB	BASIC DYNAMIC LOAD RATING LB	APPROX LIMITING SPEED RPM	Y END THICK-NESS MAX
	NOM	MIN	MAX	RING GAGE	OUTSIDE NO.		MIN	MAX				
MS52141 -17	7/8	.8755	.8764	1.1245	1-1/8	.750	.8745	.8750	3020	2470	14000	.04
MS " -18	7/8	.8755	.8764	1.1245	1-1/8	1.000	.8745	.8750	4610	3440	14000	.04
MS " -19	1	1.0005	1.0014	1.2495	1-1/4	.750	.9995	1.0000	3320	2620	12000	.04
MS " -20	1	1.0005	1.0014	1.2495	1-1/4	1.000	.9995	1.0000	5080	3640	12000	.04
MS " -21	1-1/8	1.1255	1.1264	1.3745	1-3/8	.750	1.1245	1.1250	3630	2740	11000	.04
MS " -22	1-1/8	1.1255	1.1264	1.3745	1-3/8	1.000	1.1245	1.1250	5540	3810	11000	.04
MS " -23	1-1/4	1.2505	1.2514	1.4995	1-1/2	1.000	1.2495	1.2500	6000	3990	10000	.04
MS " -24	1-1/2	1.5005	1.5016	1.8745	1-7/8	1.000	1.4995	1.5000	7760	5700	8600	.05
MS " -25	1-1/2	1.5005	1.5016	1.8745	1-7/8	1.250	1.4995	1.5000	10500	7230	8600	.05
MS " -26	1-3/4	1.7505	1.7517	2.1245	2-1/8	.750	1.7495	1.7500	5540	4230	7300	.05
MS " -27	1-3/4	1.7505	1.7517	2.1245	2-1/8	1.000	1.7495	1.7500	8330	5810	7300	.05
MS " -28	1-3/4	1.7505	1.7517	2.1245	2-1/8	1.500	1.7495	1.7500	14100	8770	7300	.05
MS " -29	1-7/8	1.8755	1.8767	2.2495	2-1/4	1.000	1.8745	1.8750	8700	6000	6800	.05
MS " -30	2	2.0006	2.0018	2.3745	2-3/8	1.000	1.9994	2.0000	9270	6190	6300	.05

MATERIAL: ROLLERS: STEEL, ALLOY OR CARBON, E50100, E51100, E52100 OR 1090 OR 1095 PER FED-STD-66 OR SAE J-404.
OUTER RING: STEEL ALLOY OR CARBON, CARBURIZING GRADE, 4620, 4720, 8620, 8720 OR 1010-1020 PER FED-STD-66.

HEAT TREATMENT: NEEDLE ROLLERS THROUGH HARDENED TO RC58 MIN.
OUTER RING: CASE HARDENED TO SURFACE HARDNESS RC58-65. CASE DEPTH, .003 MIN.

SURFACE FINISH: SURFACES MARKED $\sqrt{8}$ AND $\sqrt{40}$ SHALL HAVE MAXIMUM SURFACE ROUGHNESS AS INDICATED, PER ANSI B46.1.

PROTECTIVE COATING: PLAIN (NOT PLATED). BEARINGS TO BE COATED WITH RUST PREVENTIVE FILM BY MANUFACTURER.

LUBRICATION: BEARINGS SHALL BE FURNISHED WITHOUT LUBRICATION.

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. ROLLERS SHALL BE RETAINED BY THE CAGE.
3. BEARINGS SHALL NOT BE FURNISHED WITH ROLLER SEPARATORS.
4. THE "GO" PLUG GAGE IS THE SAME SIZE AS THE MINIMUM DIAMETER UNDER THE NEEDLE ROLLERS, AND THE "NO GO" PLUG GAGE SIZE IS .0001 INCH LARGER THAN THE MAXIMUM DIAMETER UNDER THE NEEDLE ROLLERS. INSPECTION OF THE BEARING BORE SHOULD BE MADE WITH THE BEARING PRESSED INTO A RING GAGE OF THE SIZE SHOWN IN THE RING GAGE DIAMETER COLUMN.
5. a. BEARINGS ARE INTENDED TO BE INSTALLED ON SHAFTS WHERE MAXIMUM DEFLECTION DOES NOT EXCEED .0010 INCHES PER INCH OF BEARING WIDTH.

   b. APPLICATIONS INVOLVING OSCILLATING MOTION OFTEN REQUIRE REDUCED RADIAL CLEARANCES. THIS REDUCTION IS ACCOMPLISHED BY INCREASING THE SHAFT RACEWAY DIAMETERS BY .0003 FOR BEARINGS WITH BORES 3/32 TO 3/16; .0005 FOR BEARINGS WITH BORES 1/4 TO 1 7/8; .0006 FOR BEARINGS WITH BORES 2 TO 5 1/2.
6. BEARINGS ARE INTENDED TO BE USED WITH HARDENED SHAFTS RC58-65. WHEN OPEN END BEARINGS ARE USED WITH UNHARDENED SHAFTS, BEARINGS SHOULD BE USED IN CONJUNCTION WITH RINGS, BEARING, INNER, MS17130.
7. RECOMMENDED STEEL HOUSING BORE DIMENSIONS ARE AS FOLLOWS: RING GAGE DIAMETER PLUS .0005 MINUS .0000, FOR BEARING OUTSIDE DIAMETERS THROUGH 11/32; RING GAGE DIAMETER PLUS .0010 MINUS .0000, FOR BEARING OUTSIDE DIAMETERS 7/16 THROUGH 5; RING GAGE DIAMETEP PLUS .0020 MINUS .0000, 5-1/16 AND GREATER.

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8. MOUNTING IN CONFORMANCE WITH THE RECOMMENDED SHAFT DIAMETERS AND HOUSING BORE DIAMETERS RESULT IN THE FOLLOWING DIAMETRAL CLEARANCES: .0005 TO .002, FOR BEARINGS WITH BORE DIAMETERS UP TO AND INCLUDING 3/16; .0005 TO .0029, FROM 1/4 THROUGH 1-1/4; .0005 TO .003, FROM 1-5/16 THROUGH 1-3/8; .0005 TO .0031, FROM 1-1/2 THROUGH 1-5/8; .0005 TO .0032, FROM 1-3/4 THROUGH 1-7/8; .0006 TO .0034, FOR 2; .0006 TO .0036, FROM 2-1/8 THROUGH 2-1/2; .001 TO .004, FROM 2-5/8 THROUGH 3-1/2; .001 TO .0056, FOR 5-1/2.

9. OIL HOLE SHALL BE FURNISHED IN ACCORDANCE WITH MANUFACTURER'S STANDARD PRACTICE.

10. THE "BASIC STATIC LOAD RATING" FOR A RADIAL ROLLER BEARING IS THAT STATIC RADIAL LOAD WHICH CORRESPONDS TO A COMBINED TOTAL PERMANENT DEFORMATION OF ROLLER PLUS RACEWAY, AT THE MOST HEAVILY STRESSED CONTACT, OF .0001 OF THE ROLLER DIAMETER.

11. THE "BASIC DYNAMIC LOAD RATING" FOR A RADIAL ROLLER BEARING IS THAT CALCULATED, CONSTANT, RADIAL LOAD WHICH A GROUP OF APPARENTLY IDENTICAL BEARINGS WITH STATIONARY OUTER RING CAN THEORETICALLY ENDURE FOR A RATING LIFE OF ONE MILLION REVOLUTIONS OF THE SHAFT. THE RATING LIFE IS DEFINED AS THE NUMBER OF REVOLUTIONS THAT 90% OF A GROUP OF BEARINGS WILL COMPLETE OR EXCEED BEFORE THE FIRST EVIDENCE OF FATIGUE DEVELOPS. SINCE APPLIED LOADING AS GREAT AS THE BASIC DYNAMIC LOAD RATING TENDS TO CAUSE LOCAL PLASTIC DEFORMATION OF THE ROLLING SURFACES, IT IS NOT ANTICIPATED THAT SUCH HEAVY LOADING WOULD NORMALLY BE APPLIED.

12. FOR RATING LIVES AT LOADS OTHER THAN THE BASIC DYNAMIC LOAD RATINGS, AT OPERATING CONDITIONS OF THE INNER RING ROTATING, OUTER RING STATIONARY, STEADY LOAD, UNIFORM RPM, THOROUGH LUBRICATION, 300° F MAXIMUM BEARING TEMPERATURE, SHAFT MISALINEMENT NOT TO EXCEED .0010 INCHES PER INCH OF BEARING WIDTH.

13. LIMITING SPEED CALCULATED AT 300° F MAXIMUM WITH THOROUGH LUBRICATION AT BASIC LOAD RATING.
EMPIRICAL FORMULA FOR LOAD-LIFE RELATIONSHIP:

$$L = \left|\frac{Cr}{P}\right|^{\frac{10}{3}}, \text{ WHERE}$$

L = RATING LIFE, MILLIONS OF REVOLUTIONS.
Cr = BASIC DYNAMIC LOAD RATING, LB.
P = EQUIVALENT RADIAL LOAD TO WHICH BEARING IS SUBJECTED, LB.

FOR AN APPLIED LOAD GREATER THAN ONE-HALF THE BASIC DYNAMIC LOAD RATING, THE ABOVE LOAD-LIFE RELATIONSHIP IS NOT VALID.

14. AVERAGE LIFE, THE LIFE WHICH 50% OF A GROUP OF BEARINGS WILL ATTAIN BEFORE EVIDENCE OF FATIGUE DEVELOPS, IS APPROXIMATELY FIVE (5) TIMES THE RATING LIFE.

15. THE MS PART NUMBER SHALL CONSIST OF THE MS DASH NUMBER PLUS "J" FOR OPEN END BEARING (EXAMPLE: MS52141-1J); OR PLUS "MJ" FOR CLOSED END BEARING (EXAMPLE: MS52141-1MJ). MARKING SHALL CONSIST OF THE MS PART NUMBER AND THE MANUFACTURER'S IDENTIFICATION IN ACCORDANCE WITH MIL-STD-130.

16. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.

17. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

18. THE USE OF RECYCLED MATERIALS WHICH MEET THE REQUIREMENTS OF THE APPLICABLE MATERIAL SPECIFICATION WITHOUT JEOPARDIZING THE INTENDED USE OF THE ITEM SHALL BE ENCOURAGED.

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